

SLOP1836

Human-Cybernetic Organ Interaction

This is a course about rewiring how you think so that you can converse with CCS's using their own language. We'll harness this newfound intelligence towards making art (literary, visual, and auditory)

Admin

- Lectures – 10am -> 12pm Thursdays @ the Cybernetics Space Lecture Theatre
- Labs — start next week check your timetable and get assigned to one!
- Assignments — literary piece (20%) + visual artwork (20%) + auditory piece (20%) + workshop engagement (40%) = 100%
- Getting info and help
- Course reps

Learning Outcomes

By the end of the course you will be able to:

- create and critique artworks in the mediums of visual, auditory, and literary.
- explain how CCS's work, including their capabilities as well as the scholarly literature's current understanding of their biochemistry
- utilising scholarly literature's understanding of CCS cognition, implement and evaluate different patterns for the purpose of complex reasoning.

Cortical bionic symbiote



“relating to the cerebral cortex (= the outer layer of the brain, responsible for language, thinking, creating new ideas, etc.)”

Cortical **bionic** symbiote



“having normal biological capability or performance enhanced by or as if by electronic or electromechanical devices”

Cortical bionic **symbiote**

some nonsense we don't care about as
computer scientists (save it for the
chemists)

What is a CCS?

- Techniques that can be traced back to investigations by Adamatzky (2015) into utilising properties of *Physarum Polycephalum* as a biocomputer
- matrices of bio-engineered neurons in a manner not too dissimilar from artificial neural networks
- creation of CCS's as we know them today pioneered by companies such as General Intelligence (business focused) and StandardAI (consumer focused)
- still lots of research to be done about mechanisms of CCS interaction with the brain's electrochemistry
- fun ethical question: are they conscious? -> biological, hyperintelligent

Elephant in the room

- General Intelligence grant to SLOP University \$500,000
- Would this course exist without grant? Probably not
- But the grant only covers costs of access to CCS organs, self-implant kits, nutritions only

Even though this course would not exist without them and we are grateful for their contribution, **General intelligence has no role in the course's design, content, or assessment.**

Boring stuff (legal from SLOP U)

Anything can have side effects. Sometimes they are serious, most of the time they are not.

- A study indicated increased risk of heart attack, stroke, cancer, and blood clots in the legs or lungs in those implanted with a CCS.
- The researchers stopped the study after 6 months when it was determined the risks were greater than the benefits in this group.
- Another study indicated increased risk of dementia in people aged above 22 years who had a CCS implant.
- Significant weight loss, memory loss, and inattentivity may be an undesirable result of prolonged CCS implantation

Hating CCS properly

You cannot complete the coursework without CCS implantation. In other words, leave the room, drop the course, goodbye, good luck.

Getting started with your CCS

- Using latest CCS iteration
- Still 1-2 month “Gestation” period – side effects associated with this period please read about them and consult your doctor if any arise
- Hughes suggests you implant tonight (see your self-implantation kit)! Must do it before first workshop

CCS interaction

- interact with a CCS using your thoughts (not like LLM where you use text)
- naïve approach: just think
- the approach we will implement and develop: learning to think in the language of the CCS

CCS Language

- Lots of research interest into the bionic organism's "language of thought" — how they understand and then execute (emergent behaviour)
- Crux of this course, is harnessing our current understanding of CCS language to interact more efficiently and ultimately obtain better results when using them

We can use this communication method to harness CCS to do anything but we're doing artwork

What is art?

There is no generally agreed definition of what constitutes *art*, and its interpretation has varied greatly throughout history and across cultures.

— <https://en.wikipedia.org/wiki/Art>

What is art?

- Plato (375BC)
art is imitation
- about copying something out of the real world
- the goodness of the imitation influences the goodness of the art
- Kant (1790)
a kind of representation that is purposive in itself
- representation as in something in the world
- not just descriptive, has it's own role in and of itself

What is art?

- Michel Duchamp (1910s)
art is anything
 - anything you call art
 - not just about representation, being a pretty picture, etc.
- (see more at the [Stanford Encyclopedia of Philosophy](#))

CCS & art

- Literary, auditory, and visual artworks are well-represented in CCS training sets — well supported application domain
- Useful way to practice what we learn in terms of thought communication and interfacing your body with CCS (e.g. learning painting technique)
- However, we can enhance CCS using techniques we will discuss throughout the course (such as with siphoning)